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EDUCATION

New York University, Tandon School of Engineering

MS, Computer Science, May 2026. GPA 3.67/4.0

IIIT Delhi

B.Tech, Computer Science and Applied Mathematics; minor in Economics, 2020 - 2024. CGPA 8.19/10

Coursework: Stochastic Processes, Statistical Inference, Convex Optimization, Game Theory, Econometrics, Valuation and Portfolio Management, Machine Learning

PROJECTS

trading-agent (local only) | Python, pandas, LightGBM, TabPFN, Chronos-2 2026

- Cost-aware evaluation harness (purged walk-forward, CPCV, Deflated Sharpe, PBO, random-portfolio null) over 14 models on 8 US ETFs, 2011 to 2024.
- No candidate beat vol-targeted equal weight after costs (best 0.68 Sharpe vs 0.81), so none was promoted; zero-shot Kronos called 5-day direction right 46.6%.
- Execution path shipped regardless: Alpaca and Zerodha Kite adapters, a \$100 paper account on launchd, drawdown throttle, kill switch, four gates before real money.

Non-stationary bandits | Python, NumPy, Matplotlib github.com/happyc0der/multi_armed_bandit_algorithms

- Five piecewise-stationary bandit algorithms (TS-GE, AdSwitch, M-UCB, UCB1, epsilon-greedy) behind one interface, benchmarked from 2 to 128 arms.
- Audited TS-GE against its paper and found the Beta update reversed: TS-phase regret 21,860 to 174; mandatory probing is 75.9% of its total regret.
- Made AdSwitch 59x faster (218.6 s to 3.7 s) and cut memory from 2,048 MB to 272 MB at K=128, T=1e6, bit-identical output.

Econometrics re-analysis (team of 5) | R, github.com/utkar22/Econometric-Analysis-of-Agriculture-and-Health-Across-India
fixest, data.table, Python, pandas

2023 coursework with four teammates in Utkarsh Arora's repo; the 2026 re-analysis, pipeline, models, review and CI are mine.

- Re-analysed a 2023 econometrics team project on 673 Indian districts; reproduced both tables to 6 decimals, then showed the crop indices were row-order artefacts.
- Rebuilt it as a deterministic Python and R (fixest) pipeline with state-clustered SEs and fixed effects; the original headline effects vanish, within-R2 below 0.02.
- Wrote the review, fixed 7 run-breaking bugs in the archived scripts, added GPL-3.0 and CI that reruns the pipeline and fails on any changed table.

Kaggle S6E9, EV purchases | Python, LightGBM, XGBoost, CatBoost, github.com/happyc0der/kaggle-s6e9-ev-purchases
PyTorch

- Kaggle Playground S6E9 (ROC AUC): out-of-fold 0.94648, public leaderboard 0.94649 against a leader at 0.94674; final ranking pending.
- Nested target encodings over the synthetic generator's artifacts on 668,665 rows, 20 folds; logit blend of LightGBM, CatBoost and a 15-seed MLP.
- 192 experiments judged by paired DeLong tests; label resampling showed a perfect model scores 0.9459 ± 0.0003 , so the gap left is label noise.

COMPETITIONS

- Biohub Cell Tracking, Kaggle**, 2026. Public leaderboard 0.952. Final ranking pending, competition ends 2026-09-29.
- Kaggle Playground S6E9**, 2026. Out-of-fold AUC 0.9465. Final ranking pending, competition ends 2026-09-30.
- TrafficFlowBench**, 2026. Public leaderboard 0.845 against a 0.553 baseline. In progress, ends 2026-11-06.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

ML: PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, Hugging Face transformers, vLLM, Ollama, NumPy, pandas, polars

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Streamlit, Altair, matplotlib

Tools: Git, GitHub Actions, Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale

ACTIVITIES

- Events head, Finnexia (finance club), IIIT Delhi; organised "The Critical Bid"
- Attended the first International Quantum Communication Conclave, New Delhi, 2023